

$$7. e^{i(\alpha+\beta)} = e^{i\alpha} e^{i\beta} \quad (\alpha, \beta \in \mathbb{R})$$

$$\Rightarrow C(\alpha+\beta) + iS(\alpha+\beta) = (C\alpha + iS\alpha)(C\beta + iS\beta)$$

$$= (C\alpha C\beta - S\alpha S\beta) + i(S\alpha C\beta + C\alpha S\beta)$$

$$\Rightarrow C(\alpha+\beta) = C\alpha C\beta - S\alpha S\beta$$

$$S(\alpha+\beta) = S\alpha C\beta + C\alpha S\beta$$

$$8. \text{ let } t = \tan \frac{\theta}{2} \Rightarrow \sin \theta = \frac{2t}{1+t^2}, \quad \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\text{proof: } \because \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}$$

$$\therefore \sin \theta = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2}} = \frac{2 \frac{t}{1+t^2}}{\frac{1-t^2}{1+t^2} + 1} = \frac{2t}{1+t^2}$$

$$\cos \theta = \frac{\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}}{\cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2}} = \frac{1-t^2}{1+t^2}$$

$$9. \because \cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \\ = 1 - 2 \sin^2 \frac{\theta}{2} \\ = 2 \cos^2 \frac{\theta}{2} - 1$$

$$\therefore \sin^2 \frac{\theta}{2} = \frac{1-\cos \theta}{2}$$

$$\cos^2 \frac{\theta}{2} = \frac{1+\cos \theta}{2}$$

$$4. (\tan^{-1} x)' = \frac{1}{1+x^2}$$

$$\text{proof: let } y = \theta = \tan^{-1}(x),$$

$$\therefore x = \tan \theta$$

$$\therefore \frac{dx}{dx} = \frac{dx}{d\theta} = \frac{1}{\sec^2 \theta} = \frac{1}{1+x^2} \quad (\text{from 1})$$

$$5. (\sec \theta)' = \tan \theta \sec \theta$$

$$\text{proof: } (\sec \theta)' = \left(\frac{1}{\cos \theta} \right)'$$

$$= -\frac{1}{\cos^2 \theta} (-\sin \theta)$$

$$= \tan \theta \sec \theta$$

$$6. (\ln(\sec \theta + \tan \theta))' = \sec \theta, \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

$$\text{proof: left} = \frac{1}{\sec \theta + \tan \theta} ((\sec \theta)' + (\tan \theta)') \\ = \frac{\sec \theta (\sec \theta + \tan \theta)}{\sec^2 \theta + \tan^2 \theta} = \sec \theta \quad (\text{here } y = \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}), \therefore \cos y > 0)$$

$$(\text{here } \because \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}), \therefore \sec \theta + \tan \theta > 0)$$

$$\therefore \ln(\sec \theta + \tan \theta) \text{ is defined}$$

some derivatives of trig functions

$$1. y = \tan^{-1} x \Rightarrow \sec y = \sqrt{1+x^2}$$

$$\text{proof: } \because x = \tan y$$

$$\therefore 1+x^2 = 1 + \frac{\sin^2 y}{\cos^2 y}$$

$$= \frac{1}{\cos^2 y}$$

$$= \sec^2 y$$

$$\therefore y = \theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\therefore \sec \theta > 0 \quad \therefore \sec \theta = \sqrt{1+x^2}$$

$$2. (\sin^{-1}(x))' = \frac{1}{\sqrt{1-x^2}}$$

$$\text{proof: } y = \tan^{-1} x \quad \therefore \frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+x^2}$$